Research Article

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Epigenomics

Integrated comparison of the miRNAome and mRNAome in muscles of dermatomyositis and polymyositis reveals common and specific miRNA–mRNAs

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Aim: Dermatomyositis (DM) and polymyositis (PM) are refractory systemic autoimmune diseases with unknown pathogenesis. miRNAs is an important epigenetic mechanism to regulate gene expression. **Methods:** We performed whole miRNAs analysis, transcription analysis and the association between miRNAome and mRNAome. **Results:** For transcription and miRNAs analysis, there were common and specific mRNAs and miRNAs in the muscles of DM and PM. Among them, the expression levels of miR-196a-5p and *CPM* were negatively correlated in PM, miR-193b-3p and *NECAP2* were negatively correlated in DM and PM. Protein carboxypeptidase M (CPM) plays roles in the degradation of extracellular proteins and in the migration and invasion of cancer cells, and protein NECAP2 plays roles in adaptor protein AP-1-mediated fast recycling from early endosomes. The functions of them in the pathogenesis of DM/PM need further studies. **Conclusion:** Our study identified and confirmed differentially miRNAs and mRNAs in DM and PM. Our observations have laid the groundwork for further diagnostic and mechanistic studies of DM and PM.

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Keywords: idiopathic inflammatory myopathies • miRNAs • whole-genome transcription

Idiopathic inflammatory myopathies (IIMs) are a group of heterogeneous autoimmune diseases characterized by progressive symmetric muscle weakness, elevated serum levels of muscle enzymes, electromyographic abnormalities and inflammatory infiltrates in muscle [1]. The main clinical subtypes of IIMs are classified as dermatomyositis (DM; juvenile, adult), polymyositis (PM), sporadic inclusion body myositis and immune-mediated necrotizing myopathy [2]. The pathogenesis of DM/PM is not fully understood. Both immune and nonimmune mechanisms are causal contributors to muscle weakness [3]. The immune mechanisms include immune cells, such as T cells, B cells, dendritic cells and macrophages, and their products, such as cytokines and antibodies. The nonimmune pathogenic mechanisms involve endoplasmic reticulum stress, hypoxia and autophagy [4].

In recent years, great efforts have been made to understand the etiopathogenesis of DM/PM and indicated important roles of epigenetic alterations [5,6]. Epigenetic mechanisms such as DNA methylation, histone modification and the activities of miRNA and long non-coding RNA have been widely studied [7–10].

miRNAs play important roles in a wide range of developmental and physiological cellular processes in DM/PM. The expression levels and functions of miRNAs have been widely studied in serum, peripheral blood mononuclear cells, skeletal muscle and skin of DM/PM patients. Many miRNAs, such as miR-21, miR-206 [11], miR-3907 [12], miR-1, miR-133a/b, miR-146a, miR-221, miR-222 [13] and miR-7 [14] have been shown to be aberrantly expressed in DM/PM patients and potentially contribute to its pathogenesis. The miRNAome–mRNAome interaction network is complicated *in vivo* [15]. However, most studies have focused on identifying the functions of single miRNAs or mRNAs mainly using *in vitro* experiments, which may not reflect their real *in vivo* effects. It would therefore be of substantial value to perform systematic analyses of the miRNAome and mRNAome in DM/PM,

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which may provide new insights into the pathogenesis of DM/PM in addition to important biomarkers and therapeutic targets.

In this study, we integrated analyses of the miRNAome and mRNAome in the muscles of DM/PM. First, we analyzed the gene expression profiles in DM/PM muscle tissues (NCBI GEO Database, GSE 5370, GSE 1551, GSE 39454 and GSE 3112 [16]) and identified differentially expressed genes in DM, PM or DM/PM. This analysis was then enriched by further bioinformatics analysis. Next, we analyzed the altered miRNAs in DM, PM or DM/PM, and used the TargetScan database, PicTar database, or miRDB database to select all of the predicted mRNA targets. Third, we combined the differentially expressed genes with these predicted mRNAs. Finally, we validated these findings for differentially expressed miRNAs and mRNAs using the muscle tissues of DM/PM and human skeletal muscle myoblasts (HSkMM) that were stimulated with DM/PM serum or IFN-β.

Materials & methods

Patients & controls

We studied nine DM patients, nine PM patients and nine normal controls (NC). All patients met the European Neuromuscular Centre (ENMC) 2003 criteria and Bohan/Peter diagnostic criteria for DM and PM [17–19]. Diagnostic muscle biopsies were performed in all the enrolled patients, which showed $CD8^+$ T cells invade non-necrotic muscle fibers, the expression of MHC-I and without rimmed vacuoles in muscle fibers. The details of the patients and controls are shown in Supplementary Table 1. Before muscle biopsies surgery, all patients only take glucocorticoids for less than one week, without using any immunosuppressors. This study was approved by the Ethics Committee of Central South University (No. 201212074), and was carried out according to the medical research regulations of China. Informed consent was obtained from all patients.

Biopsy specimens & cell culture

Muscle biopsy specimens were obtained from the bicipital muscles of patients and controls, and immediately frozen in isopentane cooled with liquid nitrogen, as previously described [20]. HSkMM were purchased from Sciencell (USA) and cultured at 37°C in an atmosphere containing 5% CO_2 in Skeletal Muscle Cell Medium (SMCM, USA). Cells of passages 3–5 were used in the experiments. The cells were used when they had reached 80–90% confluence.

Treatment with DM/PM serum or IFN-β

HSkMM were cultured in serum-free SMCM (starvation medium) for 24 h and then stimulated with 20% DM/PM serum for 24 h, the serum samples were from three DM and three PM, who were first onset and given prednisone for less than 4 days without any other immunosuppressor. HSkMM were cultured for 24 h in serum-free SMCM (starvation medium), and then stimulated with IFN-β (100 U/ml) for 48 h [21].

mRNA microarray data processing

Raw data for GSE 5370, GSE 1551, GSE 39454 and GSE 3112 were downloaded from the Gene Expression Omnibus (www.ncbi.nlm.nih.gov/gds/) and used for $\log^2$ transformation and quantile normalization. R was used to perform a statistical test to determine whether mRNAs were differentially expressed among DM, PM and normal control samples. Differentially expressed mRNAs were defined as those with an adjusted p-value <0.05 and a fold change >1.2 [22,23]. After analysis, we determined the differently expressed mRNAs in DM or PM specifically, and the common genes in DM and PM.

miRNA microarray data processing & target prediction

The raw data for miRNA expression was downloaded from the raw data supplement by Georgantas *et al.* [24]. After quantile normalization, R was used to perform a statistical test to determine whether miRNAs were differentially expressed among DM, PM and normal control samples. Differentially expressed miRNAs were defined as those with an adjusted p-value <0.05 and fold change >1.2. The predicted target genes of these altered miRNAs were obtained using the TargetScan, PicTar or miRDB database.

Functional classification/gene network analyses

The gene function, biological networks and canonical pathways were analyzed using the Database for Annotation, Visualization and Integrated Discovery (DAVID) Functional Annotation Tool (https://david.ncifcrf.gov/) [22,25].

Cytoscape 3.5.1 was used to integrate information related to the biomolecular interaction networks between these miRNAs and mRNAs.

Validation using RNA isolation & quantitative reverse transcription PCR

As previously described [26], total RNA was isolated from muscle tissues and HSkMM using Trizol (Invitrogen Life Technologies, CA, USA) according to the manufacturer's instructions. For mRNAs, cDNA was prepared using the Reverse Transcription System (Promega, WI, USA). The expression of related genes was measured using gene-specific primers (shown in Supplementary Table 2) with SYBR Green (SYBR Premix Ex Taq RT-PCR kit, Takara, Japan) and the 7500 real-time PCR system analyzer (Applied Biosystems, CA, USA). GAPDH expression was used as the endogenous control to normalize the sample data. For miRNAs, the miScript reverse transcription kit (Qiagen, Germany) and miScript SYBR Green PCR kit (Qiagen) were used to measure the expression levels of selected miRNAs in a model 7500 real-time PCR system analyzer (Applied Biosystems). miRNA-specific primers purchased from RiboBio (Shanghai, China) and the miScript Universal Primer (Qiagen) were used. The expression of U6B small nuclear RNA (RNU6B) was used as the endogenous control to normalize the sample data as described previously [27]. Relative expression levels were calculated using the $2^{-\Delta Ct}$ method.

Statistical analysis

GraphPad Prism software was used for all statistical analyses (not including the microarray data). Numerical variables with a normal distribution were compared using unpaired t-tests. Data with a non-normal distribution were compared using the Mann–Whitney U test. All data are expressed as the mean ± SEM. $p < 0.05$ was considered to indicate statistical significance.

Results

Identification of differentially expressed mRNAs in DM

We reanalyzed the mRNA expression data in the muscle tissues of DM from two microarray platforms (www.ncbi.nlm.nih.gov/geo/GSE accession number GSE5370, GSE1551). GSE5370 contains five DM patients and four NC, while GSE1551 contains 13 DM patients and ten NC. We identified 1842 genes that were significantly upregulated and 1627 genes that were significantly downregulated in DM compared with NC in GSE5370, and 645 genes that were significantly upregulated and 510 genes that were significantly downregulated in DM compared with NC in GSE1551. A total of 364 genes (including 266 upregulated and 98 downregulated genes) were found to be commonly expressed both in GSE5370 and GSE1551, which may represent the most significantly altered genes in DM (adjusted $p < 0.05$ and fold change >1.2) (Supplementary Figure 1A). Moreover, we performed an exploratory two-dimensional (2D) hierarchical clustering of the 364 genes, which resulted in separate clusters indicating a substantial difference between DM and NC (Supplementary Figure 1B). To investigate the categories of common dysregulated genes, we performed DAVID functional annotation analysis [22]. Pathway analysis suggested that the differentially expressed genes in DM were involved in herpes simplex infection, phagosomes, Epstein–Barr virus infection, the NF-kappa B signaling pathway and the Toll-like receptor signaling pathway, among others (Supplementary Figure 1C). Furthermore, these genes were significantly enriched in GO biological processes, including the type I interferon signaling pathway, response to virus, defense response to virus, innate immune response and negative regulation of viral genome replication, among others (Supplementary Figure 1D).

Identification of differentially expressed mRNAs in PM

Similarly, we reanalyzed the raw data from GSE3112 and GSE39454, which contains six PM and 11 NC, eight PM and five NC, respectively. There were 2881 genes significantly upregulated and 4175 genes significantly downregulated in PM in GSE39454, while 1993 genes were significantly upregulated and 1945 genes significantly downregulated in PM in GSE3112. A total of 1447 genes (including 943 upregulated and 504 downregulated genes) were commonly expressed both in GSE3112 and GSE39454, which may represent the most significantly altered genes in PM (adjusted $p < 0.05$ and fold change > 1.2) (Supplementary Figure 2A). Exploratory 2D hierarchical clustering of the 1447 genes revealed dramatic differences between PM and NC (Supplementary Figure 2B). Pathway analysis of the common differentially expressed genes demonstrated an enrichment of genes in pathways involved in phagosome, antigen processing and presentation, *Staphylococcus aureus* infection, viral myocarditis and allograft rejection, among others (Supplementary Figure 2C). The functions of these genes were related with

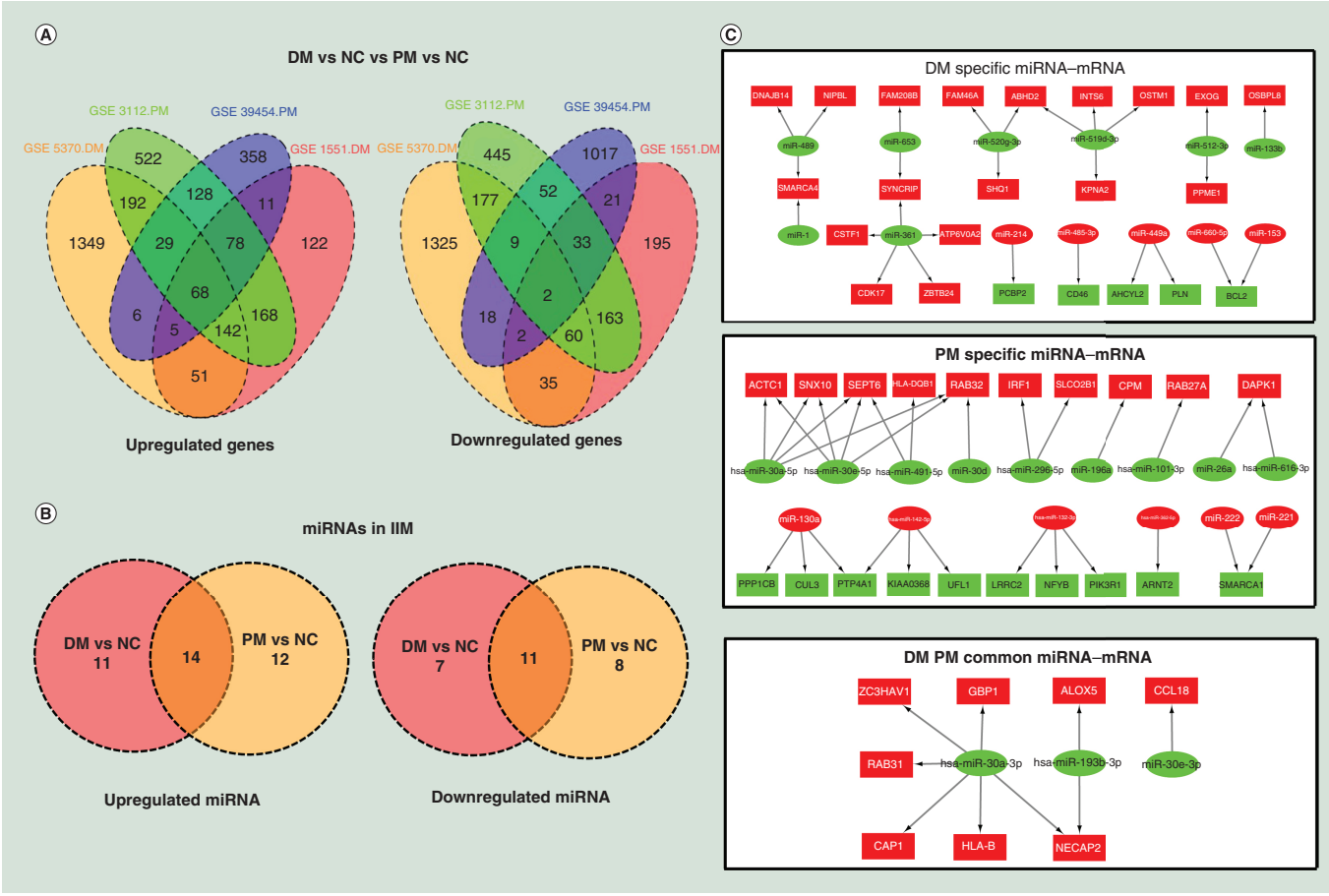

Figure 1. Identification of differentially expressed miRNAs and their target genes. (A) Venn diagram of the differential gene expression analysis in DM and PM. **(B)** Venn diagram of the differential miRNAs expression analysis in DM and PM. **(C)** Networks were combined with data for the miRNAs and their predicted targets. Nodes represent dysregulated miRNAs (circle) and genes (square) in the pathways. Node colors (red for upregulation and green for downregulation) represent fold changes in the corresponding miRNAs or genes.
DM: Dermatomyositis; NC: Normal controls; PM: Polymyositis.

IFN-γ-mediated signaling pathway, immune response, type I interferon signaling pathway, inflammatory response and defense response to virus, among others (Supplementary Figure 2D).

DM and PM have common biological functions such as immune response, IFN-γ-mediated signaling pathway, innate immune response, defense response to virus and type I interferon signaling pathway. They also have specific functions, such as response to IFN-α, response to virus, negative regulation of viral genome replication, response to IFN-β and B cell receptor signaling pathway in DM, in contrast to signal transduction, inflammatory response, leukocyte migration, antigen processing and presentation and regulation of immune response in PM (Supplementary Figure 1D & 2D).

Identification of differentially expressed mRNAs & miRNAs in DM & PM

We analyzed the gene expression data (GSE5370, GSE1551, GSE3112 and GSE39454) together, and found that 51 genes were specifically upregulated in DM, 128 genes were specifically upregulated in PM and 68 genes were commonly upregulated in DM and PM, while 35 genes were specifically downregulated in DM, 52 genes were specifically downregulated in PM and two genes were commonly downregulated in DM and PM (Figure 1A). Analysis of the miRNA data [28] showed that 11 miRNAs were specifically upregulated in DM, 12 miRNAs were specifically upregulated in PM and 14 miRNAs were commonly upregulated in DM and PM. Seven miRNAs were specifically downregulated in DM, eight miRNAs were specifically downregulated in PM and 11 miRNAs were commonly downregulated in DM and PM (Figure 1B). We combined these differentially expressed genes with the previously predicted miRNA target genes and we matched 51 gene pairs that resulted in 65 miRNA–mRNA pairs,

among which 27 miRNA–mRNA pairs were specific in DM, 29 miRNA–mRNA pairs were specific in PM and nine miRNA–mRNA pairs were common in DM and PM (Figure 1C).

Validation of miRNAs & mRNAs that were differentially expressed in DM/PM muscle tissues

To further confirm that these miRNAs and mRNAs were differentially expressed, the upregulated and downregulated miRNAs and their 51 target genes were selected for real-time PCR detection in the muscle tissues of DM/PM. In DM, there were no significant differences in miRNAs levels, and the target genes *FAM208B*, *OSBPL8*, *OSTM1*, *NIPBL*, *SHQ1*, *CDK17* and *INTS6* were upregulated (Figure 2A). In PM, miR-196a-5p, miR-30d-5p, miR-30a-5p and miR-101-3p were downregulated, and the target genes *DAPK1*, *SNX10*, *CPM*, *ACTC1* and *SCLO2B1* were upregulated (Figure 2B). miR-193b-3p was downregulated both in DM and PM, and the target genes *HLA-B*, *NECAP2* and *GBP1* were upregulated (Figure 2C). In summary, the expression levels of miR-196a-5p and *CPM* were negatively correlated in PM. The expression levels of miR-193b-3p and *NECAP2* were negatively correlated both in DM and PM. We also detected the specificity of these altered miRNAs and mRNAs, miR-196a-5p was specifically downregulated in PM; *FAM208B*, *OSBPL8*, *OSTM1*, *NIPBL*, *SHQ1*, *CDK17* and *INTS6* were specifically upregulated in DM; and *DAPK1*, *SNX10*, *CPM*, *AC TC 1* and *SCLO2B1* were specifically upregulated in PM (Figure 2D).

Validation of miRNAs & mRNAs that were differentially expressed in HSkMM after stimulation with DM/PM serum or IFN-β

To investigate the effect of DM or PM serum on the expression of miRNA–mRNA pairs, we stimulated normal HSkMM with 20% DM or PM serum for 24 h. The expression levels of *FAM208B*, *OSBPL8*, *OSTM1*, *CDK17*, *I NTS6* and *AHCYL2* were upregulated when stimulated with DM serum (Figure 3A). After stimulation with PM serum, the expression levels of miR-132-3p, miR-221-3p, miR130a-3p, miR-196a-5p, miR-616-3p, miR-491-5p, miR-101-3p, miR-30a-5p, miR-30e-5p, miR-30d-5p, miR-26a-5p and miR-296-5p were upregulated, and their target genes *ARNT2*, *KIAA0368*, *PIK3R1*, *NFYB*, *PPP1CB*, *RAB32*, *DAPK1*, *SNX10*, *CPM* and *RAB27A* were also upregulated (Figure 3B). *HLA-B*, *GBP1* and *RAB31* were upregulated both by DM and PM serum (Figure 3C).

Because recent microarrays studies have shown that IFN-induced genes are among the most differentially expressed genes in IIM muscles [29,30], we used IFN-β (100 U/ml, 48 h) to stimulate normal HSkMM. We found that the expression levels of miR-520g-3p and miR-519-3p were higher and miR-1-3p was lower in normal HSkMM that were stimulated with IFN-β (Figure 4A). The expression levels of *CD46*, *OSBPL8*, *OSTM1*, *SHQ1*, *CDK17*, *INTS6, FAM46A, EXOG, K PNA2, CS TF1, DNA JB14, PPP1CB, CU L3, NFYB , CPM , CAP 1, HLA -B, GBP1* and *RAB31* were elevated (Figure 4B).

In summary, after stimulation with serum or IFN-β, the altered miRNAs in HSkMM were not consistent with these in the muscles of patients. However, the changes in *OSBPL8, OSTM1, CDK17, INTS6, CPM, HLA-B* and *GBP1* were consistent with those in the muscle of patients.

Discussion

Using genome-level miRNAs and gene transcription analysis, we identified differentially expressed miRNAs and genes in DM and PM respectively, among which common miRNAs and gene expression alterations were observed in both DM and PM, as well as specific miRNA and gene expression data to DM or PM. These miRNAs and genes may be useful to develop biomarkers for diagnostic or prognostic purposes. We integrated the miRNA data with gene expression profiles and found that miRNAs may have a regulatory effect on many of the differentially expressed genes in DM and PM, enabling us to investigate the complexity of gene dysregulation in the context of this disease. To our knowledge, this is the first study to evaluate the role of miRNAs together with gene expression in DM and PM in a comprehensive manner.

In our study, all of the enrolled patients were first onset or relapsed after having stopped the treatments by personal decision. Diagnostic muscle biopsies were performed in all patients. Before muscle biopsies, patients were only given glucocorticoids for less than 1 week, without any immunosuppressants to reduce the influence of treatments options on the expression of miRNAs and mRNAs. In our study, the serum levels of CK and LDH were significantly higher in PM patients than DM patients, which were in accordance with previous studies, as PM patients always have more severe muscle disease [31,32]. As for the treatment during whole hospitalization, DM patients had higher utilization rates of thalidomide and hydroxychloroquine, which were proven to have positive effects on the cutaneous manifestations of DM [33,34].

Figure 2. Validation of the data showing differential expression of miRNAs and mRNAs in idiopathic inflammatory myopathy muscles. Real-time PCR was performed to analyze the levels of **(A)** miRNAs and mRNAs in DM, **(B)** miRNAs and mRNAs in PM(**C & D**) and miRNAs and mRNAs both in DM and PM. All experiments were repeated at least three-times. Data are presented as $2^{(-\Delta CT)}$ relative to the levels of *RNU6B* or *GAPDH*.
*,#Indicates $p < 0.05$ versus NC.
DM: Dermatomyositis; NC: Normal controls; PM: Polymyositis.

Previous studies have demonstrated that the expression of some miRNAs are altered in DM and PM and participate in disease pathogenesis. In this study, we identified novel miRNAs that were specific or common to DM and PM. The miR-196a-5p was specifically downregulated in PM, and miR-193b-3p, miR-30d-5p, miR-30a-5p and miR-101-3p were downregulated both in DM and PM. In previous studies, Robert *et al.* found that miR-193b was also downregulated both in DM and PM, which was consistent with the results from this study. However, in the

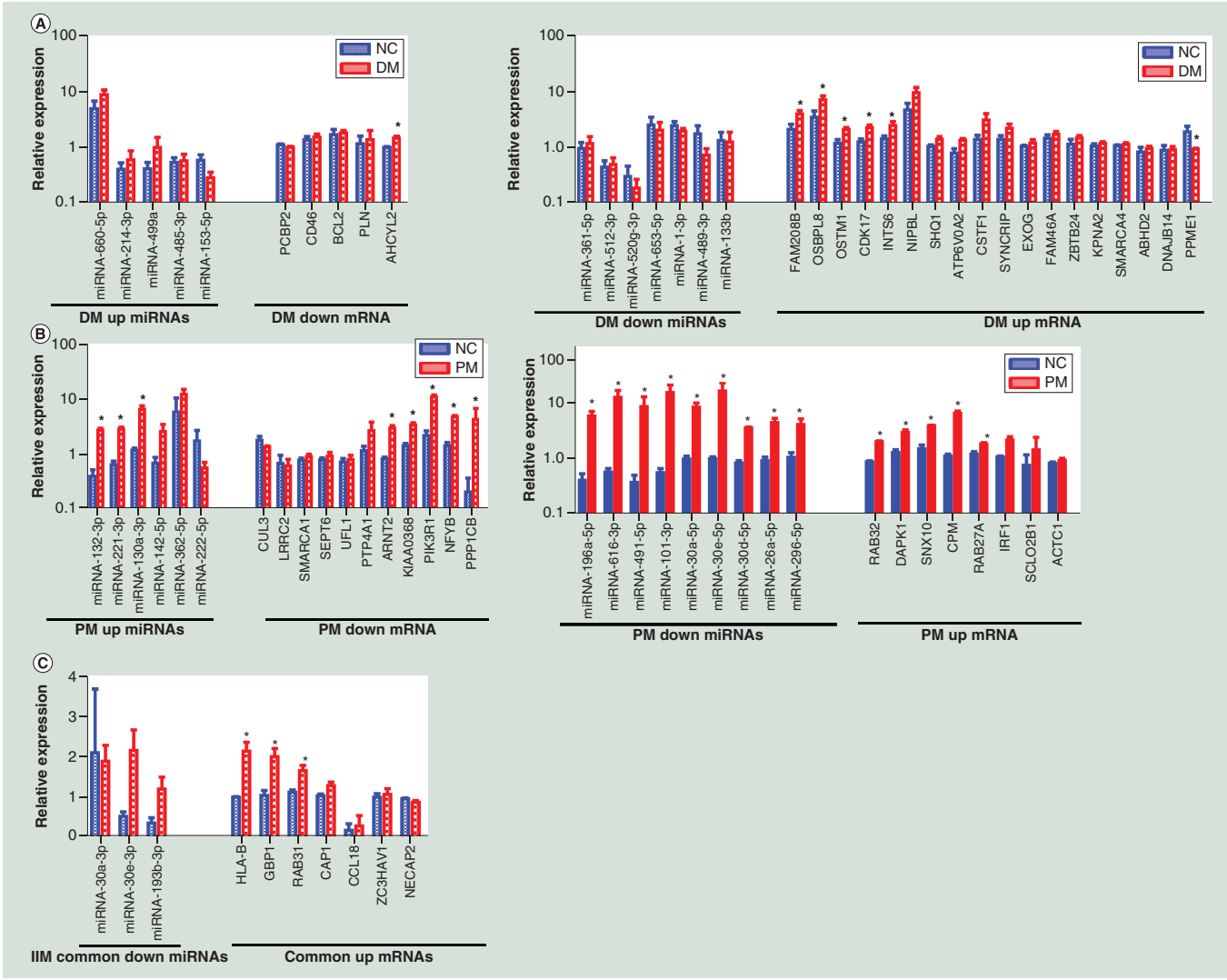

Figure 3. miRNAs and mRNAs were differentially expressed in human skeletal muscle myoblasts that were stimulated with idiopathic inflammatory myopathy serum. Real-time PCR was performed to analyze the levels of **(A)** miRNAs and mRNAs in human skeletal muscle myoblasts with DM serum stimulation, **(B)** miRNAs and mRNAs in HSkMM with PM serum stimulation **(C)** and miRNAs and mRNAs in human skeletal muscle myoblasts with DM and PM serum stimulation. All experiments were repeated at least three-times. Data are presented as $2^{(-\Delta CT)}$ relative to the levels of *RNU6B* or *GAPDH*.
*Indicates $p < 0.05$ versus NC.
DM: Dermatomyositis; NC: Normal controls; PM: Polymyositis.

Robert *et al.* study, miR-101 and miR-30a-5p were downregulated both in DM and PM, miR-30d was specifically downregulated in DM and miR-196a had no significant alterations, which were in contrast to the results garnered here [24]. In other studies, miR-101, miR-193b, miR-30a and miR-30d were downregulated in myositis, miR-196a had no changes, but this study took myositis as a whole instead of dividing the subgroups of DM, PM and included body myositis [13]. The causes of these differences could be due to: first, race, disease activity or the disease course of each of the enrolled patients differing. Second, the sites of sampling were different (quadriceps or biceps). Third, the degree of inflammation was not consistent between different samples.

The miRNA–mRNAs pairs miR-196a-5p-*CPM* were identified in PM and miR-193b-3p-*NECAP2* in DM and PM. Protein carboxypeptidase M (CPM) is an extracellular glycosylphosphatidyl-inositol-anchored membrane glycoprotein, playing roles in the control of peptide hormone and growth factor activity at the cell surface, in the membrane-localized degradation of extracellular proteins and in the migration and invasion of cancer cells [35,36]. Protein NECAP2 is a member of the adaptin-ear-binding coat-associated protein family, and contributes in adaptor

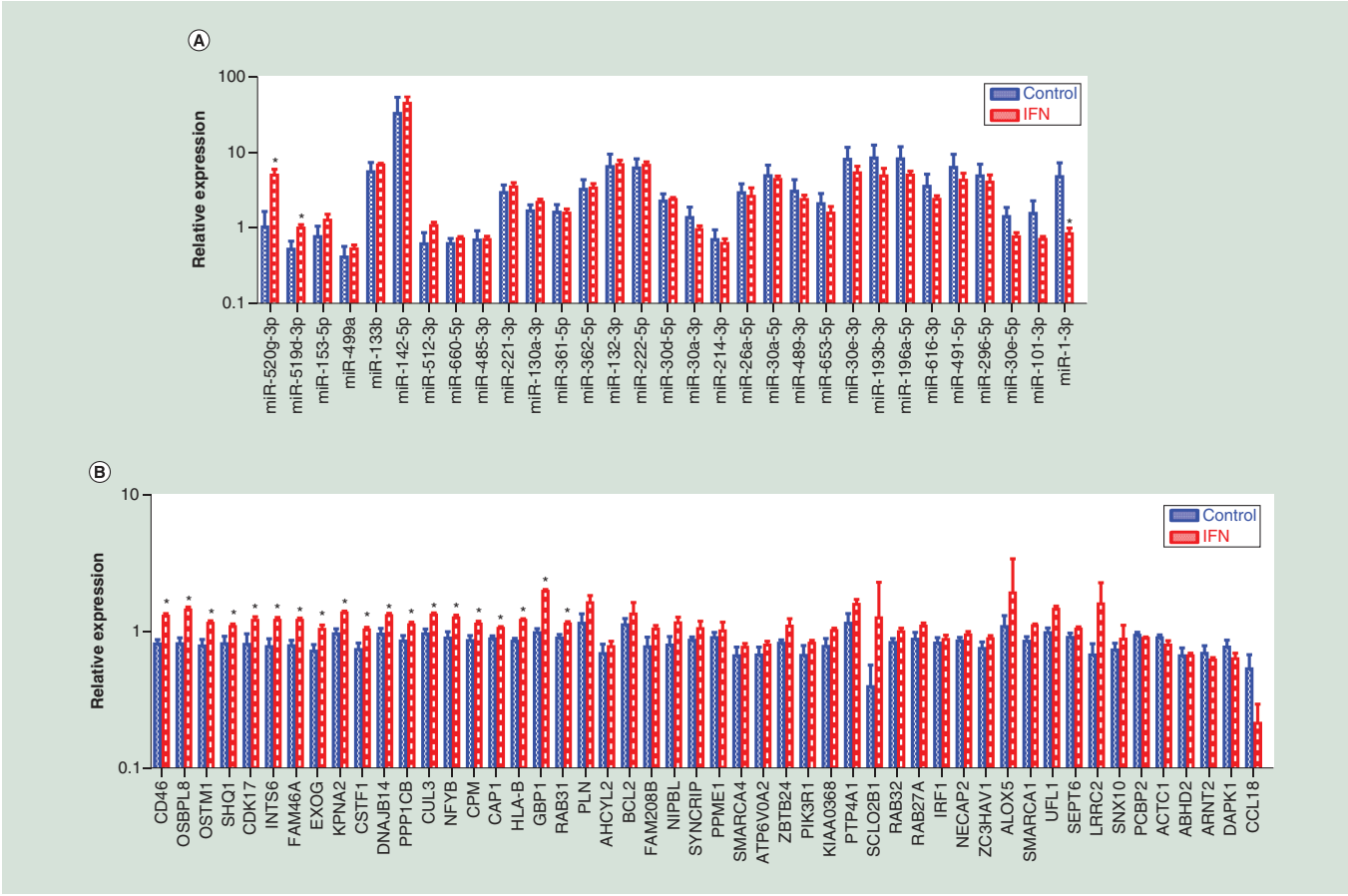

Figure 4. miRNAs and mRNAs were differentially expressed in human skeletal muscle myoblasts that were stimulated with IFN-β. Real-time PCR was performed to analyze the levels of **(A)** mRNAs in human skeletal muscle myoblasts with IFN-β stimulation, **(B)** and miRNAs in human skeletal muscle myoblasts with polymyositis serum stimulation. All experiments were repeated at least three-times. Data are presented as $2^{(-\Delta CT)}$ relative to the levels of *RNU6B* or *GAPDH*.
*Indicates $p < 0.05$ versus normal controls.

protein AP-1-mediated fast recycling from early endosomes [37]. In the muscle of IIM patients and HSkMM stimulated with serum or IFN-β, *OSBPL8, OSTM1, CDK17, INTS6, CPM, HLA-B* and *GBP1* expression were upregulated. *H LA-B* is regarded as a genetic risk factor in IIMs [38] and is associated with anti-Jo-1 and anti-SRP autoantibodies [10,39]. *GBP1* is an IFN-induced gene, is not regulated in IIM peripheral blood mononuclear cells and is associated with disease activity [40,41]. There have been no reports about the other six genes associated with IIMs. This is the first study to describe the upregulation of these genes in IIMs and potentially play an important role in the underlying mechanisms. These novel findings regarding miRNA and mRNA expression profiles provide clues about the mechanisms underlying the pathology of IIMs and therefore merit further functional analysis.

Our observations lay the groundwork for further diagnostic and mechanistic studies of DM/PM that could lead to improvements in detection, patient classification and therapy. However, this study also has some limitations. First, the miRNAome and mRNAome data were not collected from the same muscle tissues, were potentially influenced by the variation of patients, diseases status and the manipulation process. However, to minimize error, we used the mRNAome data from two centers, and the raw data were reanalyzed using strict normalization and statistical methods. Second, the miRNAome data were obtained from TaqMan Low-Density Arrays, which only contain 365 miRNAs and thus are relatively small. Consequently, this analysis may not be sufficient for combination with the mRNAome data. More samples for real-time PCR validation were used, which could compensate for this deficiency. Third, further functional analysis must be conducted to confirm the miRNA–mRNAs pairs and novel genes in DM/PM.

Future perspective

Currently, the pathogenesis of IIM is largely unknown. As an important epigenetic modification which may repress more than 60% of mammalian protein-coding genes, miRNAs play critical roles in disease etiology. Here, we first identified specific and common altered miRNAs in DM and PM. In depth study of the role of these miRNAs will help us better understand the mechanisms of IIM. Due to their remarkable stability and reliable detection, miRNAs could be ideal biomarkers for disease diagnosis and prognosis. According to the diagnostic criteria of IIM, sometimes it's not easy to distinguish between DM and tumor associated DM, PM, sporadic inclusion body myositis and immune-mediated necrotizing myopathy. Specific miRNAs would be valuable for the subgroups diagnosis. miRNAs that are associated with disease activities, muscle damage or regeneration would be useful for determining IIM prognosis. In addition, these miRNAs could also represent novel treatment targets. Therefore, investigating the mechanism of miRNAs in IIM will be of paramount importance in the coming years, and if these miRNAs are promised as biomarkers or drug targets, more studies will be needed in future.

Summary points

- Dermatomyositis (DM) and polymyositis (PM) have common biological functions such as immune response, IFN-γ-mediated signaling pathway, type I interferon signaling pathway, etc. They also have specific functions respectively.
- DM and PM have specific and common altered mRNAs.
- DM and PM have specific and common altered miRNAs. In the muscle tissues of patients, we identified miR-196a-5p-*CPM* in PM, miR-193b-3p- *NECAP2* both in DM and PM.
- In muscle myoblasts stimulated with DM/PM serum or IFN-β, the altered miRNAs were not consistent with these in the muscles of patients. However, the changes in *OSBPL8, OSTM1, CDK17, INTS6, CPM, HLA-B* and *GBP1* were consistent.

Supplementary data

To view the supplementary data that accompany this paper please visit the journal website at: https://www.futuremedicine.com/doi/suppl/10.2217/epi-2018-0064

Author contributions

S Gao participated in study design, performed data analysis and drafted the manuscript. H Zhang participated in mRNA microarray data processing, X Zuo participated in miRNA microarray data processing and target prediction, Y Xiao participated in sample collection, D Liu participated in the cell culture and treatment, H Zhang participated in the study design, data analysis and drafted the manuscript. H Luo participated in the study design, and revised and finalized the manuscript. All authors read and approved the final manuscript.

Financial & competing interests disclosure

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